

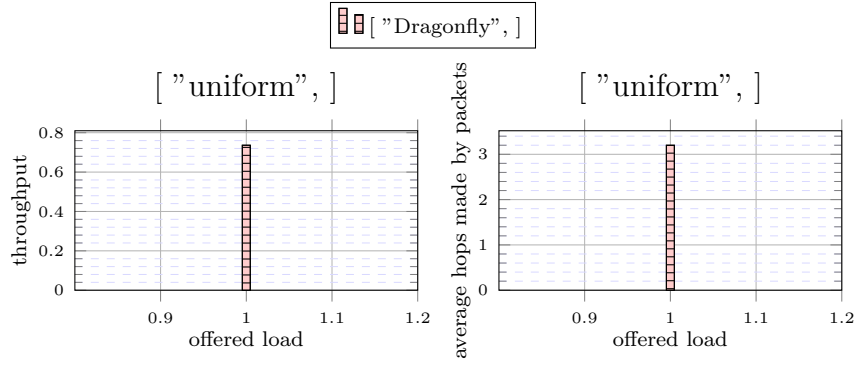


Figure 1: X: ["uniform",]

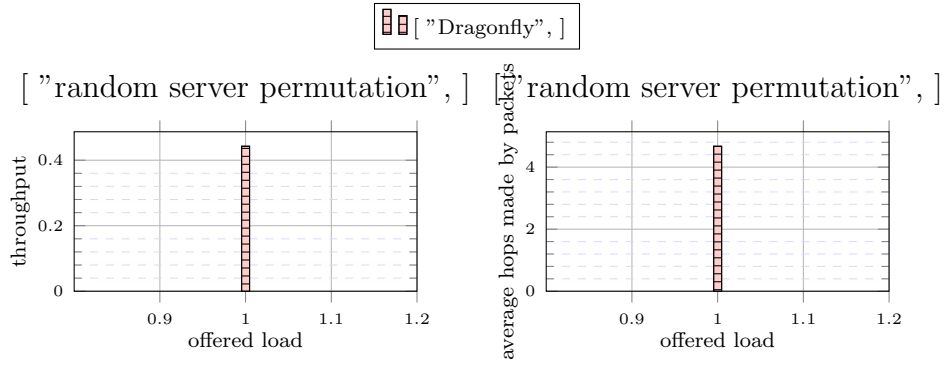


Figure 2: X: ["random server permutation",]

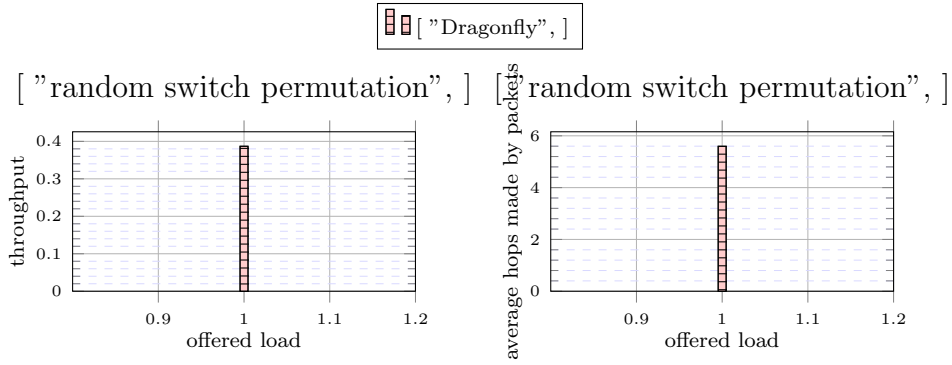


Figure 3: X: ["random switch permutation",]

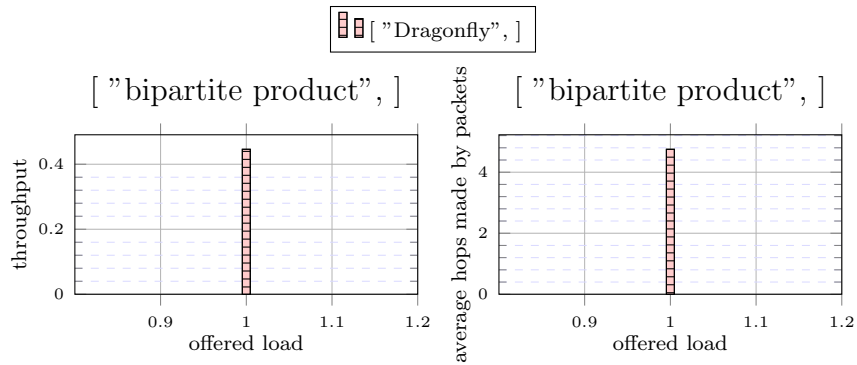


Figure 4: X: ["bipartite product",]

The following versions used in the simulations.

- heads/master-dirty-65e4f7db651d154c0abb3e3d0a691be7d6d92d82(0.6.3)

```
Configuration{
  random_seed: ![ 42, 935, 128643 ],
  warmup: 20000,
  measured: 50000,
  general_frequency_divisor: 2,
  topology: topologia![
    Dragonfly{ global_ports_per_router: 8, servers_per_router: 8, global_arrangement: Palmtree, group_size: 16, number_of_groups: 129, legend_name: "Dragonfly" }],
  traffic: HomogeneousTraffic{
    pattern: ![
      Uniform{ legend_name: "uniform" },
      RandomPermutation{ legend_name: "random server permutation" },
      CartesianTransform{
        sides: [ 8, 2064 ],
        patterns: [ Identity, RandomPermutation ],
        legend_name: "random switch permutation" },
      Product{
        block_size: 8256,
        block_pattern: Uniform,
        global_pattern: CartesianTransform{
          sides: [ 2 ],
          shift: [ 1 ],
          legend_name: "" },
        legend_name: "bipartite product" }],
    servers: 16512,
    load: ![ 0.5 ],
    message_size: 16,
    maximum_packet_size: 16,
    router: InputOutput{
      virtual_channels: topologia![ 4 ],
      virtual_channel_policies: topologia![
        [
          MapLabel{
            label_to_policy: [
              Chain{
                policies: [
                  OccupancyFunction{ label_coefficient: 0, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true, use_neighbour_space: true, aggregate: false }]],
              Chain{
                policies: [
                  AverageOccupancyFunction{
                    label_coefficient: 0,
                    occupancy_coefficient: 2,
                    product_coefficient: 0,
                    constant_coefficient: 16,
                    use_internal_space: true,
                    use_neighbour_space: true,
                    exclude_minimal_ports: true,
                    virtual_channels: [ 0 ] }]]],
                LowestLabel,
                EnforceFlowControl,
                Random]],
            allocator: Random,
            buffer_size: 128,
            bubble: false,
            flit_size: 16,
            intransit_priority: false,
            allow_request_busy_port: true,
            output_buffer_size: 64,
            crossbar_frequency_divisor: 1,
            crossbar_delay: 2,
            routing: topologia![
              Sum{
                policy: TryBoth,
                first_routing: ChannelMap{
                  routing: AscendantChannelsWithLinkClass{
                    routing: WeighedShortest{
                      class_weight: [ 1, 100 ],
                      bases: [ 1, 1 ] },
                    map: [
                      [ 0 ],
                      [ 1 ] ] },
                  second_routing: Valiant4Dragonfly{
                    first: ChannelMap{
                      routing: AscendantChannelsWithLinkClass{
                        routing: WeighedShortest{
                          class_weight: [ 1, 100 ],
                          bases: [ 1, 1 ] },
                        map: [
                          [ 0 ],
                          [ 1 ] ] },
                      second: ChannelMap{
                        routing: AscendantChannelsWithLinkClass{
                          routing: WeighedShortest{
                            class_weight: [ 1, 100 ],
                            bases: [ 1, 1 ] },
                            map: [
                              [ 2 ],
                              [ 3 ] ] }],
                        first_allowed_virtual_channels: [ 0, 1, 2, 3 ],
                        second_allowed_virtual_channels: [ 0, 1, 2, 3 ],
                        second_extra_label: 1,
                        legend_name: "ugal - minFirst" }],
                    link_classes: [
                      LinkClass{ delay: 2 },
                      LinkClass{ delay: 2 },
                      LinkClass{ delay: 2 } ],
                    launch_configurations: [
                      Slurm{ job_pack_size: 1, time: "3-00:00:00" }]]
```